

Gerardo Treviño

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EXPERIENCE

Spacelab MX

Monterrey, NL

Computer Vision Lead Engineer

Jun. 2023 – Present

- Oversee the design, development, and implementation of cutting-edge algorithms for real-time autonomous systems, utilizing deep learning models like DETR, SAM with frameworks such as PyTorch and NVIDIA DeepStream.
- Coordinate with cross-functional teams, including software developers and data scientists, to integrate computer vision technologies into security and defense systems, utilizing JavaScript and AWS.
- Ensure the successful deployment of vision systems in real time by conducting rigorous testing, optimizing algorithms for performance and accuracy, and implementing robust validation procedures, while maintaining high standards.

Computer Vision Engineer

May 2022 – June, 2023

- Designed and developed computer vision algorithms comparing models and methods for autonomous drones, enabling efficient object identification using YOLOv8.
- Developed scripts for both backend and frontend of applications to integrate and effectively use vision technologies, allowing the implementation of algorithms and the visualization of results in interactive and user-friendly interfaces.
- Implemented image processing techniques to enhance the quality and clarity of visual data, including designing and creating custom programs to apply filters, color corrections, noise removal, and other transformations to facilitate information extraction from images.

CIMA

Querétaro, QRO

Computer Vision Research

June 2020 – April 2022

- Implemented computer vision algorithms on embedded systems, optimizing performance and computational efficiency for real-time applications using YOLOv3 Tiny.
- Collected, processed, and analyzed data for predictive modeling in data science to obtain relevant insights and generate reports that drive decision-making.

INDUTELSA

Saltillo, COA

Industrial Application Developer

March 2019 – November 2019

- Developed hardware, devices, and software to set up workstations for clients in C#, ensuring specific user needs.
- Implemented devices such as HMI and PLC from Allen Bradley/Siemens/Turk, RFID, and Andon systems for workstation and plant operation.

PROJECTS

Skyprotector

- Surveillance system designed for industrial sites, enabling early detection of intrusions and quick response to emergency situations using a combination of drone technologies.

EDUCATION

M.Sc. in Artificial Intelligence

Jun. 2020 – Dec. 2021

Autonomous University of Querétaro

Querétaro, MX

B.Sc. in Electronics

Aug. 2015 – Dec. 2018

Autonomous University of Coahuila

Coahuila, MX

SKILLS

Languages : Python, JavaScript, HTML/CSS, SQL (PostgreSQL, MySQL), C#, C++

Tools : Pytorch, Docker, AWS, Figma, Notion, Jira, GitHub, Arduino, Opencv